ADDIS ABABA UNIVERSITY

SCHOOL OF BIOLOGICAL SCIENCES

DEPARTMENT OF MICROBIAL SCIENCES AND GENETICS

MASTERS OF SCIENCE IN BIOINFORMATICS

Advanced Programming Practical Assignment I

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Task 1: Environment Setup

1.1 Installing: Python, R, Pearl and VS Code

```
keti@keti-Atlas-M15-I5W303: ~
(base) keti@keti-Atlas-M15-I5W303:~$ # 1. Install Python 3.x
sudo apt update
sudo apt install python3 python3-pip -y

# 2. Install Perl
sudo apt install perl -y

# 3. Install R
sudo apt install r-base -y

# 4. Install VS Code (text editor)
sudo snap install --classic code
[sudo] password for keti:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://et.archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://et.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 https://cloud.r-project.org/bin/linux/ubuntu noble-cran40/ InRelease [3,63
1 B]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,62
0 kB]
Get:6 http://et.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:7 http://et.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,9
18 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [259
```

```
keti@keti-Atlas-M15-I5W303: ~
python3 is already the newest version (3.12.3-0ubuntu2.1).
python3-pip is already the newest version (24.0+dfsg-1ubuntu1.3).
The following packages were automatically installed and are no longer required:
  libgl1-amber-dri libglapi-mesa libllvm17t64
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 325 not upgraded.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
perl is already the newest version (5.38.2-3.2ubuntu0.2).
The following packages were automatically installed and are no longer required:
  libgl1-amber-dri libglapi-mesa libllvm17t64
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 325 not upgraded.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
r-base is already the newest version (4.5.3-1.2404.0).
The following packages were automatically installed and are no longer required:
  libgl1-amber-dri libglapi-mesa libllvm17t64
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 325 not upgraded.
snap "code" is already installed, see 'snap help refresh'
(base) keti@keti-Atlas-M15-I5W303:~$
```

1.2 Verifying Installation.

```
keti@keti-Atlas-M15-I5W303: ~
(base) keti@keti-Atlas-M15-I5W303:~$ python --version
Python 3.13.9
(base) keti@keti-Atlas-M15-I5W303:~$ R --version
R version 4.5.3 (2026-03-11) -- "Reassured Reassurer"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under the terms of the
GNU General Public License versions 2 or 3.
For more information about these matters see
https://www.gnu.org/licenses/.

(base) keti@keti-Atlas-M15-I5W303:~$ perl --version

This is perl 5, version 38, subversion 2 (v5.38.2) built for x86_64-linux-gnu-th
read-multi
(with 51 registered patches, see perl -V for more detail)

Copyright 1987-2023, Larry Wall

Perl may be copied only under the terms of either the Artistic License or the
GNU General Public License, which may be found in the Perl 5 source kit.
```

1.3 Short Explanations

Python: is an interpreted high-level programming language used in bioinformatics for sequence analysis, data manipulation, and implementing biological algorithms.

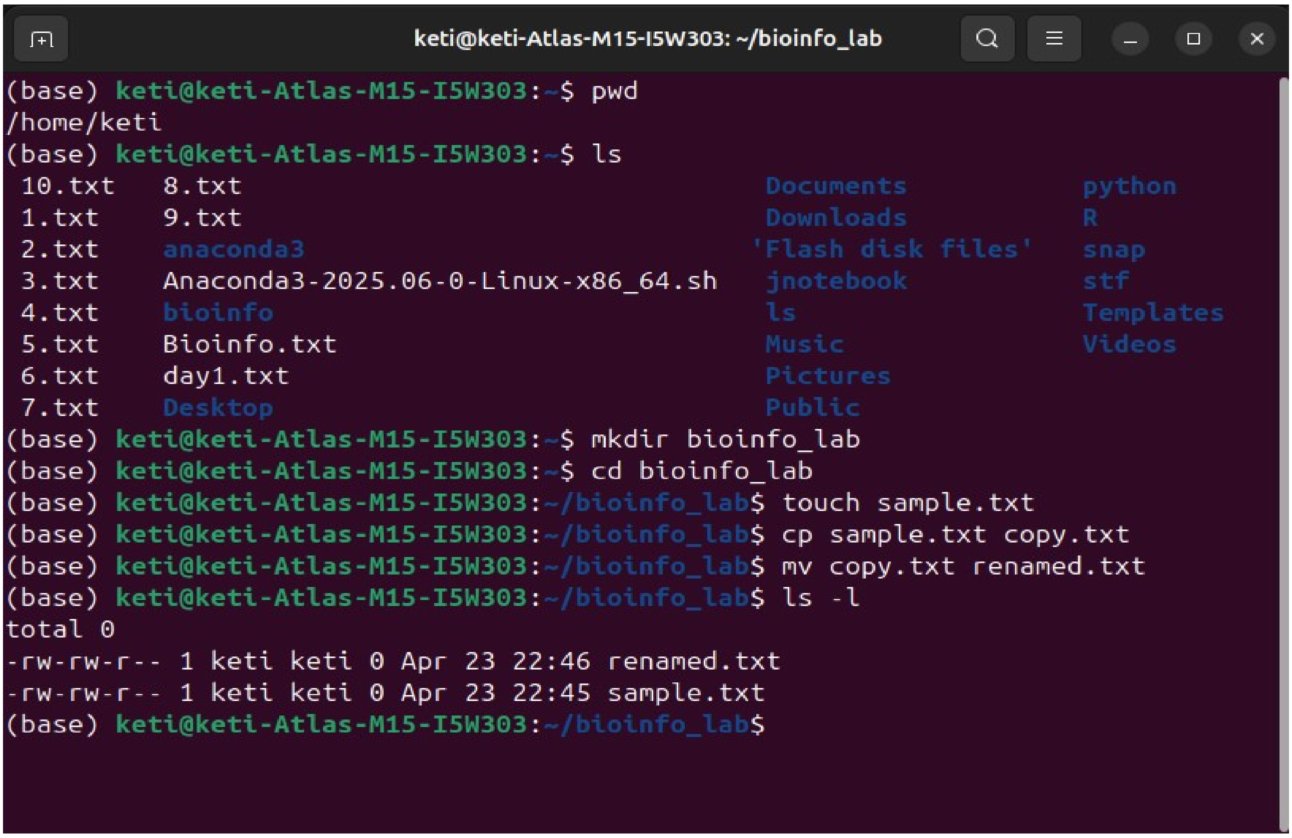
R: is a statistical computing language widely used in bioinformatics for genomic data analysis, creating publication-quality visualizations, and running Bioconductor packages for high-throughput sequencing data analysis.

Perl: is a text-processing language essential for bioinformatics, particularly for parsing FASTA files, GenBank records, and BLAST output using regex patterns.

VS Code: is a lightweight source code editor with integrated terminal and extensions for Python, Perl, and R, making it ideal for bioinformatics scripting.

Task 2: Linux File System Navigation

2.1 Command Outputs



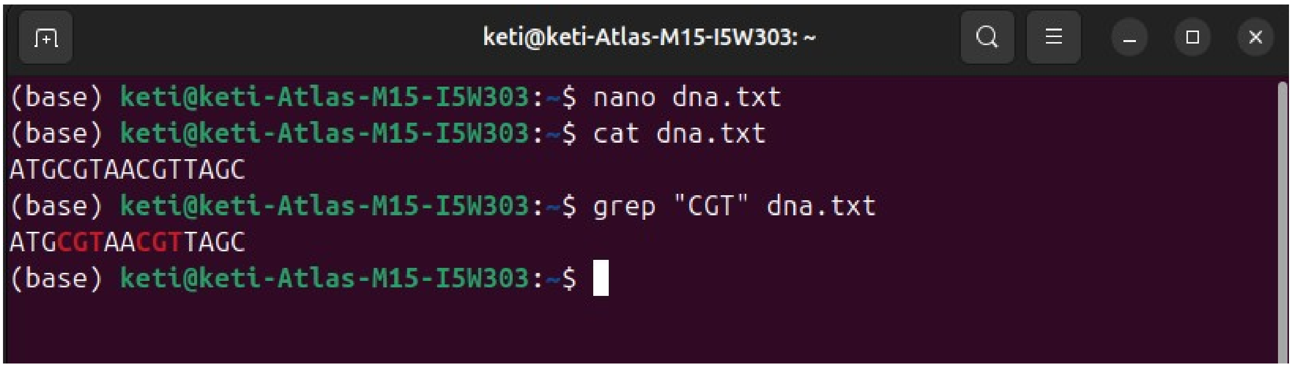
2.2 Answer to Questions

a. pwd: stands for "Print Working Directory". It displays the absolute path of the current directory you are in within the filesystem, showing exactly where you are located in the directory tree.

b. Difference between cp and mv: cp (copy) duplicates a file, creating a new file while keeping the original file intact. mv (move) relocates or renames a file, removing it from its original location. After mv, the original file no longer exists at the source; after cp, both original and copy exist.

Task 3: File Handling & Text Processing

3.1 Command Outputs



3.2 Explanation of Results

a. grep (Global Regular Expression Print): is a command-line tool that searches for text patterns within files. It reads each line of a file and prints only the lines that contain the specified search pattern.

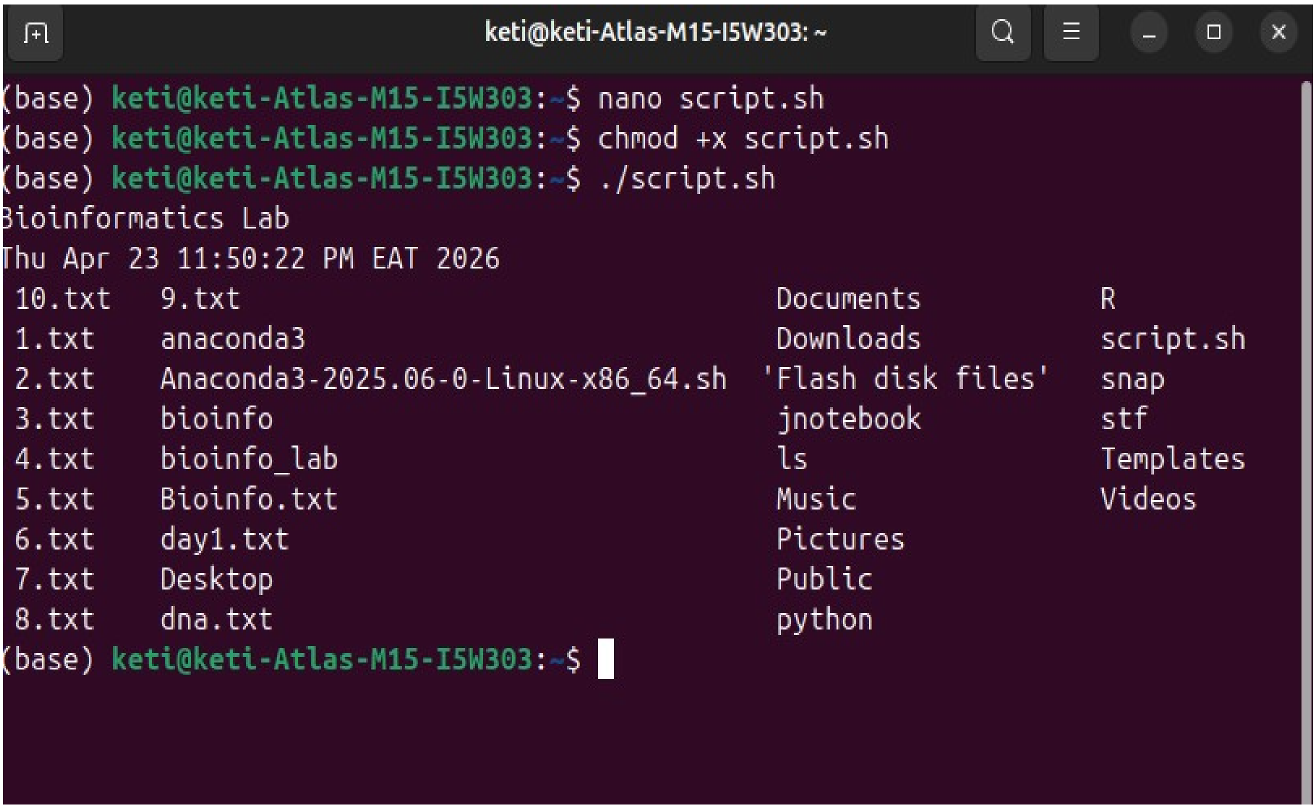
b. The output is "ATGCGTAACGTTAGC " - this entire line was printed because it contains the pattern "CGT" somewhere within it. The grep command successfully located and displayed the matching line.

The `cat` command displayed the contents of dna.txt as "ATGCGTAACGTTAGC". When `grep "CGT" dna.txt` was executed, it scanned the file and found the pattern "CGT" within the sequence. The command then output the entire line containing this match, which appeared as "ATGCGTAACGTTAGC ". This demonstrates grep's basic functionality of pattern matching and line extraction.

Task 4: Shell Scripting

4.1 Script File

4.2 Output Screenshot



4.3 Explanation of each Command

a. #!/bin/bash: This is the shebang line that specifies the Bash interpreter located at /bin/bash should execute the script. It must be the first line of any bash script.

b. echo "Bioinformatics Lab": The echo command prints the string "Bioinformatics Lab" to standard output (terminal). It acts as a header or label for the script's output.

c. date: The date command displays the current system date and time, including day, month, date, time, timezone, and year. It's useful for time stamping script executions.

d. ls: The ls command lists all files and directories in the current working directory. It shows the contents of the location where the script is run.

e. chmod +x script.sh: The chmod(change mode) command adds execute permission (+x) to script.sh, making it executable. Without this, the script cannot be run directly.

f. ./script.sh: This executes the script from the current directory (./). The shell runs each command in the script sequentially, producing the visible output.

Task 5: Simple Bioinformatics Task

5.1 Commands Used:

```
echo "ATGCTGACGTAGCTAGCTAGGCTA" > dna_sequence.txt
nano dna_sequence.txt
cat dna_sequence.txt
echo -n "ATGCTGACGTAGCTAGCTAGGCTA" | wc -c
grep -o "A" dna_sequence.txt | wc -l
```

5.2 Output

5.3 Interpretation

The DNA sequence "ATGCTGACGTAGCTAGCTAGGCTA" is 24 nucleotides long. The `wc -c` command confirmed the length as 24 characters when the newline is excluded using `echo -n`. The `grep -o "A" | wc -l` pipeline revealed 6 adenine (A) bases in the sequence, representing 25% of the total nucleotides (6/24 = 0.25). This demonstrates basic sequence analysis techniques including length calculation and nucleotide composition analysis, which are essential for bioinformatics tasks such as GC content calculation and primer design.